In words, any N point signal, $x[i]$, can be created by adding $N/2 + 1$ cosine waves and $N/2 + 1$ sine waves. The amplitudes of the cosine and sine waves are held in the arrays $Im\,\bar{X}[k]$ and $Re\,\bar{X}[k]$, respectively. The synthesis equation multiplies these amplitudes by the basis functions to create a set of scaled sine and cosine waves. Adding the scaled sine and cosine waves produces the time domain signal, $x[i]$.

In Eq. 8-2, the arrays are called $Im\,\bar{X}[k]$ and $Re\,\bar{X}[k]$, rather than $Im\,X[k]$ and $Re\,X[k]$. This is because the *amplitudes needed for synthesis* (called in this discussion: $Im\,\bar{X}[k]$ and $Re\,\bar{X}[k]$), are slightly different from the *frequency domain of a signal* (denoted by: $Im\,X[k]$ and $Re\,X[k]$). This is the scaling factor issue we referred to earlier. Although the conversion is only a simple normalization, it is a common bug in computer programs. Look out for it! In equation form, the conversion between the two is given by:

EQUATIONS 8-3
Conversion between the sinusoidal amplitudes and the frequency domain values. In these equations, $Re\,X[k]$ and $Im\,\bar{X}[k]$ hold the amplitudes of the cosine and sine waves needed for synthesis, while $Re\,X[k]$ and $Im\,X[k]$ hold the real and imaginary parts of the frequency domain. As usual, N is the number of points in the time domain signal, and k is an index that runs from 0 to $N/2$.

$$Re\,\bar{X}[k] = \frac{Re\,X[k]}{N/2}$$

$$Im\,\bar{X}[k] = -\frac{Im\,X[k]}{N/2}$$

except for two special cases:

$$Re\,\bar{X}[0] = \frac{Re\,X[0]}{N}$$

$$Re\,\bar{X}[N/2] = \frac{Re\,X[N/2]}{N}$$

Suppose you are given a frequency domain representation, and asked to synthesize the corresponding time domain signal. To start, you must find the amplitudes of the sine and cosine waves. In other words, given $Im\,X[k]$ and $Re\,X[k]$, you must find $Im\,\bar{X}[k]$ and $Re\,\bar{X}[k]$. Equation 8-3 shows this in a mathematical form. To do this in a computer program, three actions must be taken. First, divide all the values in the frequency domain by $N/2$. Second, change the sign of all the imaginary values. Third, divide the first and last samples in the real part, $Re\,X[0]$ and $Re\,X[N/2]$, by two. This provides the amplitudes needed for the synthesis described by Eq. 8-2. Taken together, Eqs. 8-2 and 8-3 *define* the inverse DFT.

The entire Inverse DFT is shown in the computer program listed in Table 8-1. There are two ways that the synthesis (Eq. 8-2) can be programmed, and both are shown. In the first method, each of the scaled sinusoids are generated one at a time and added to an accumulation array, which ends up becoming the time domain signal. In the second method, each sample in the time domain signal is calculated one at a time, as the sum of all the